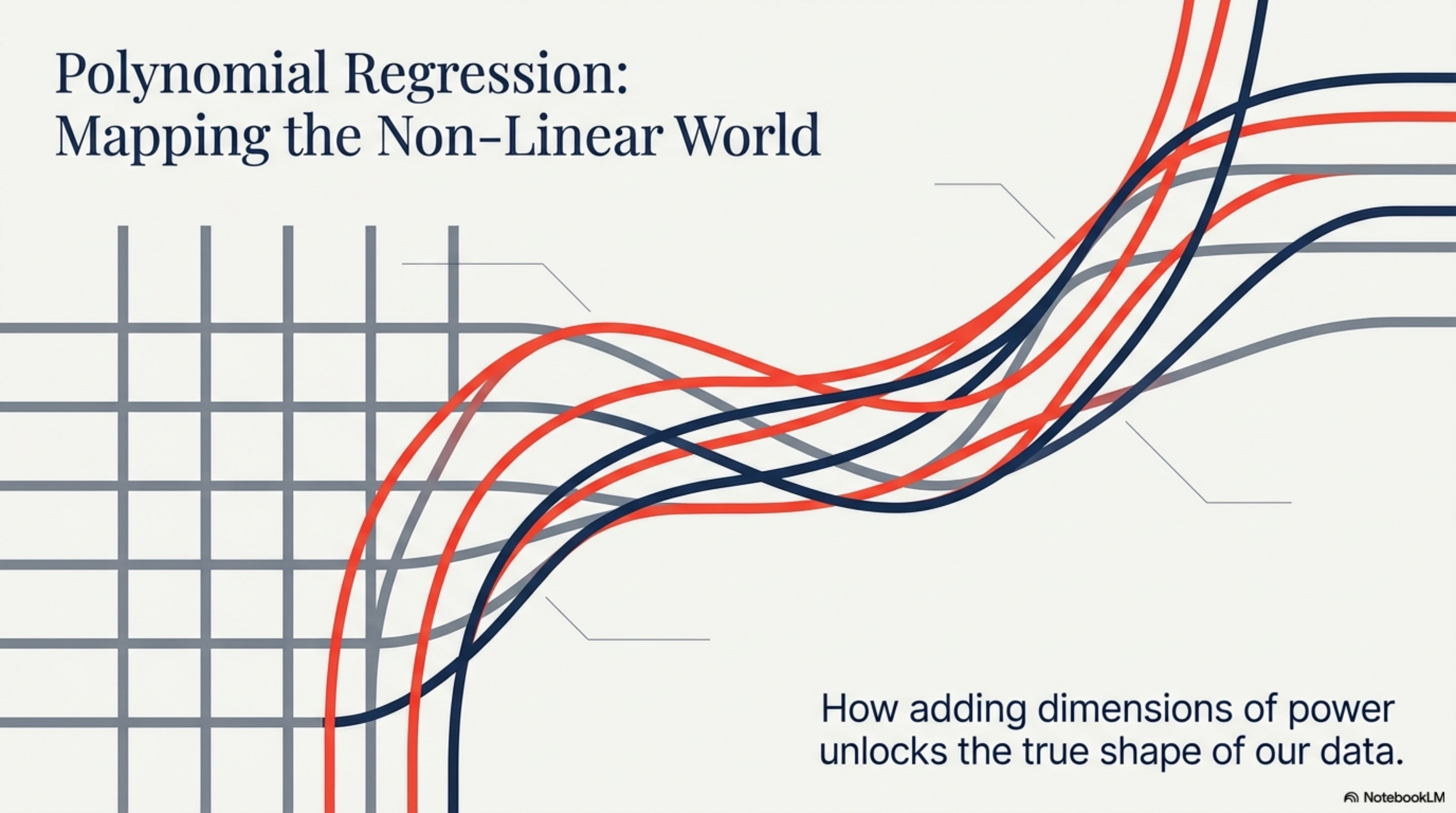


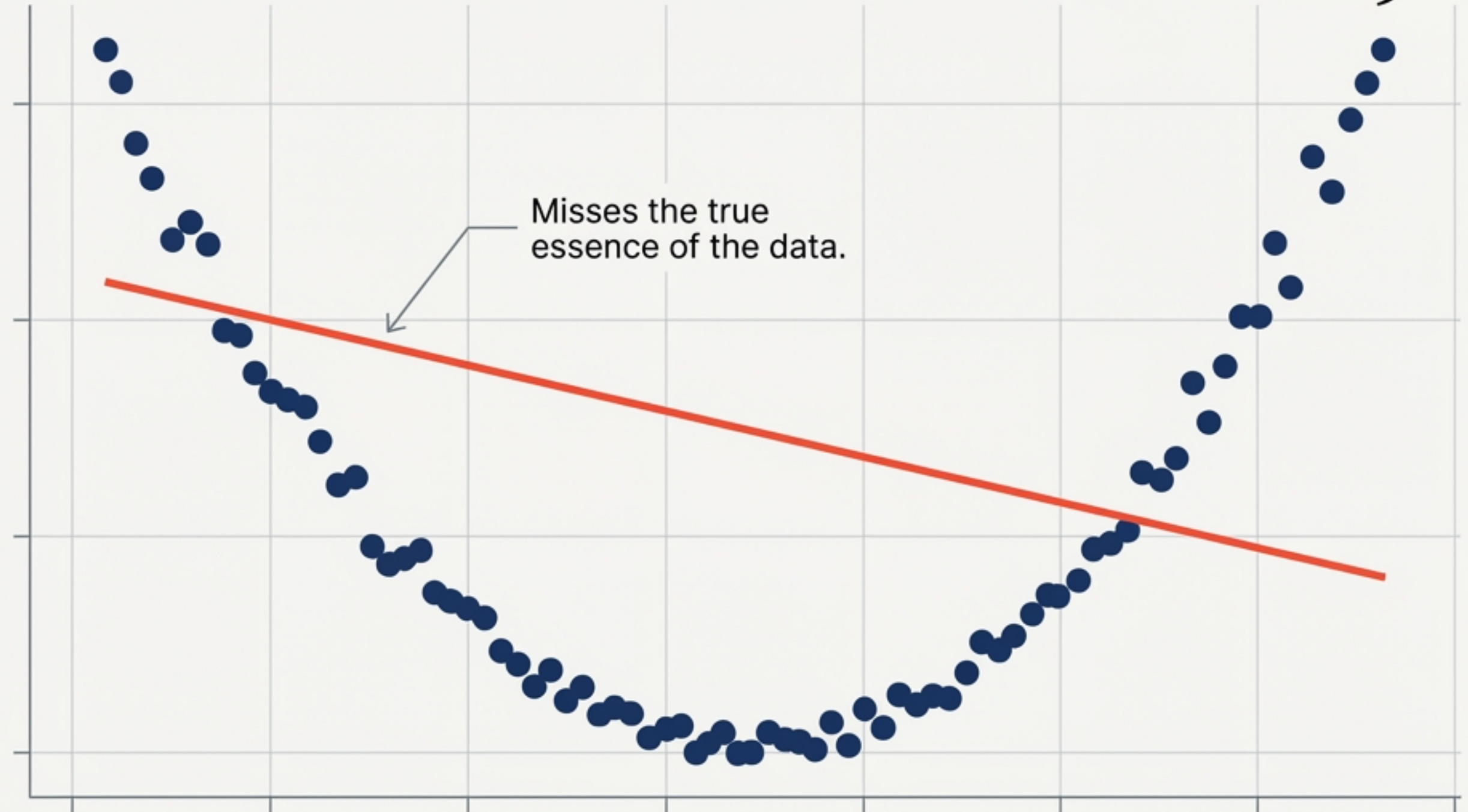
Polynomial Regression: Mapping the Non-Linear World



How adding dimensions of power
unlocks the true shape of our data.

The Limits of the Straight Line

$$R^2 = 0.192$$



Forcing a rigid straight line onto non-linear data results in massive error margins and poor predictive power.

The Polynomial Shift

$$y = \beta_0 + \beta_1 x$$

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$



By simply squaring the input feature, we give the algorithm the mechanical flexibility to bend. The data remains one-dimensional, but its mathematical representation expands.

The “Linear” Paradox

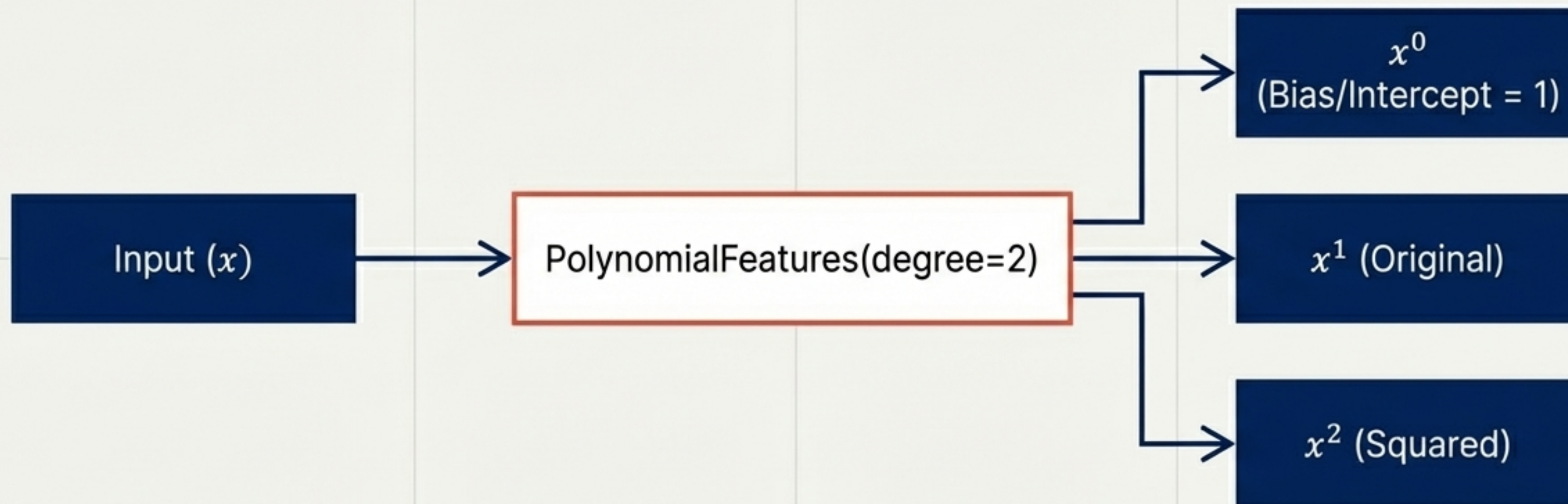
Why is this still called Linear Regression when the graph is curved?

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$



The “Linear” refers to the relationship between the unknown parameters (the β coefficients), not the x variables. Because we are still solving for linear weights, the underlying machine learning engine doesn't change.

The Transformation Pipeline



We do not change the model; we change the data. The algorithm automatically generates new features by calculating the powers of the original input.

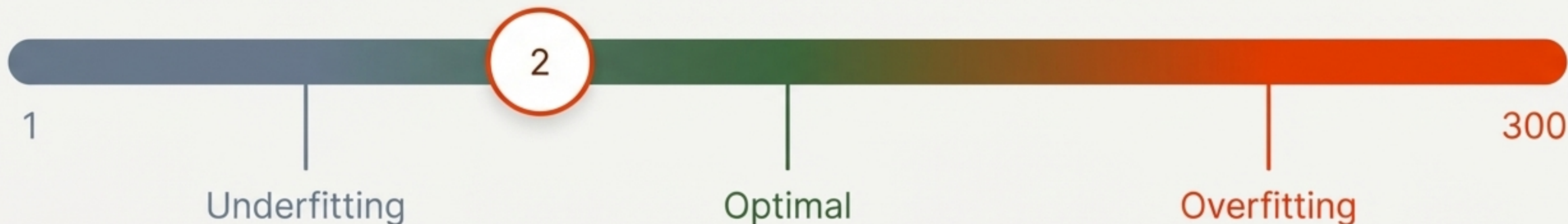
The Perfect Fit

$$R^2 = 0.817$$



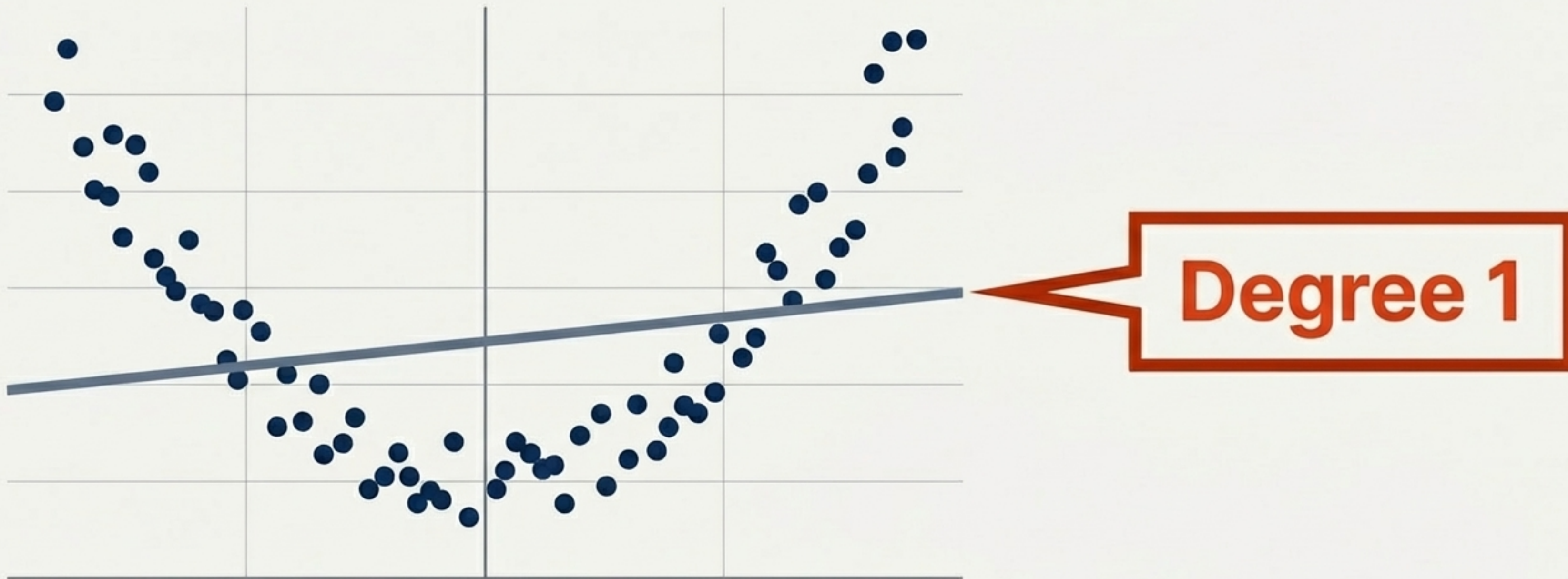
The model successfully discovers the hidden coefficients. It is not perfect due to natural data noise, but it captures the exact mathematical essence of the true pattern.

The Dilemma of Degrees



The highest risk in Polynomial Regression is deciding the Degree—the maximum power we allow the model to calculate. Choose poorly, and the model breaks down completely.

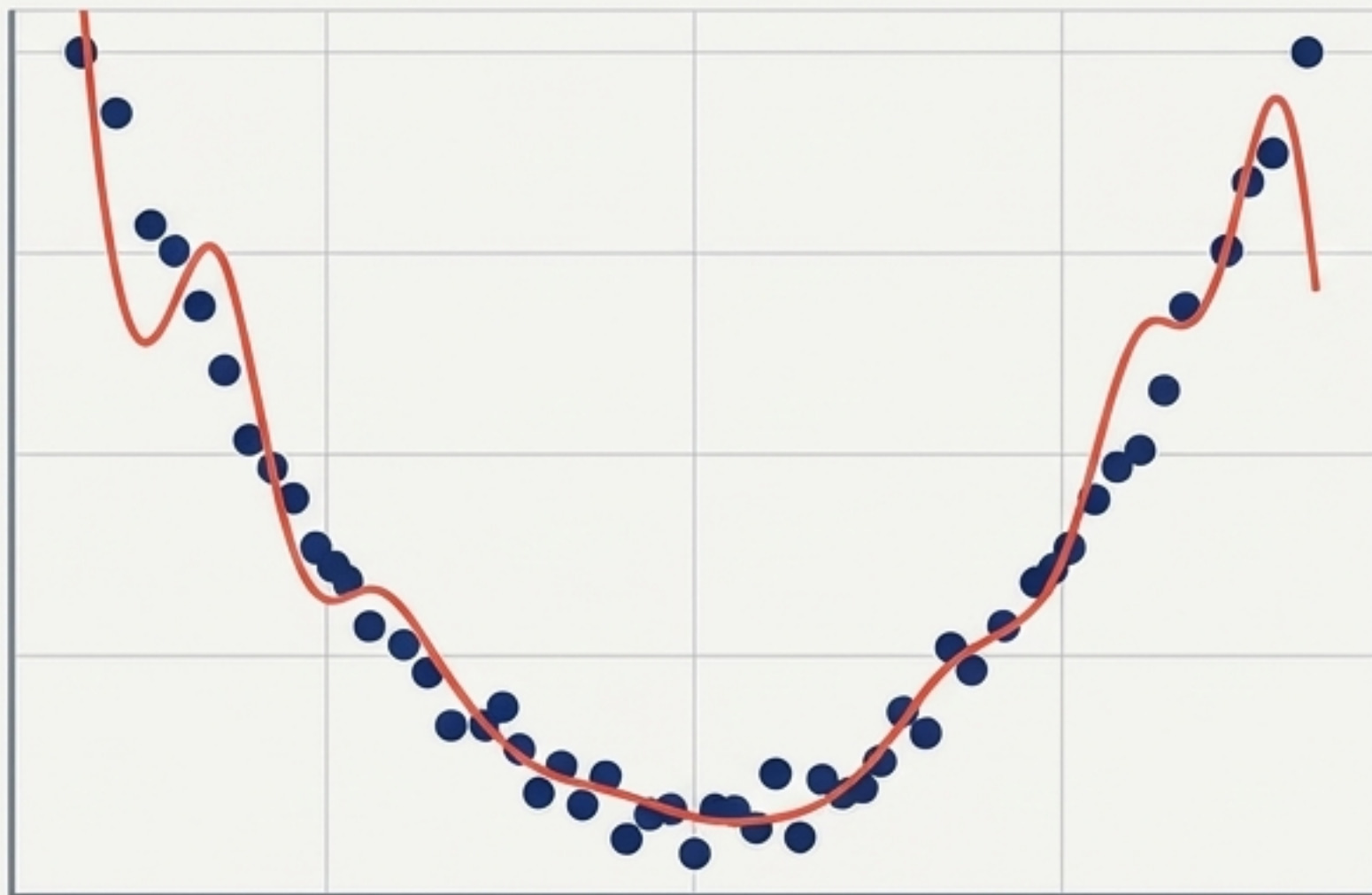
Underfitting: The Too-Simple Model



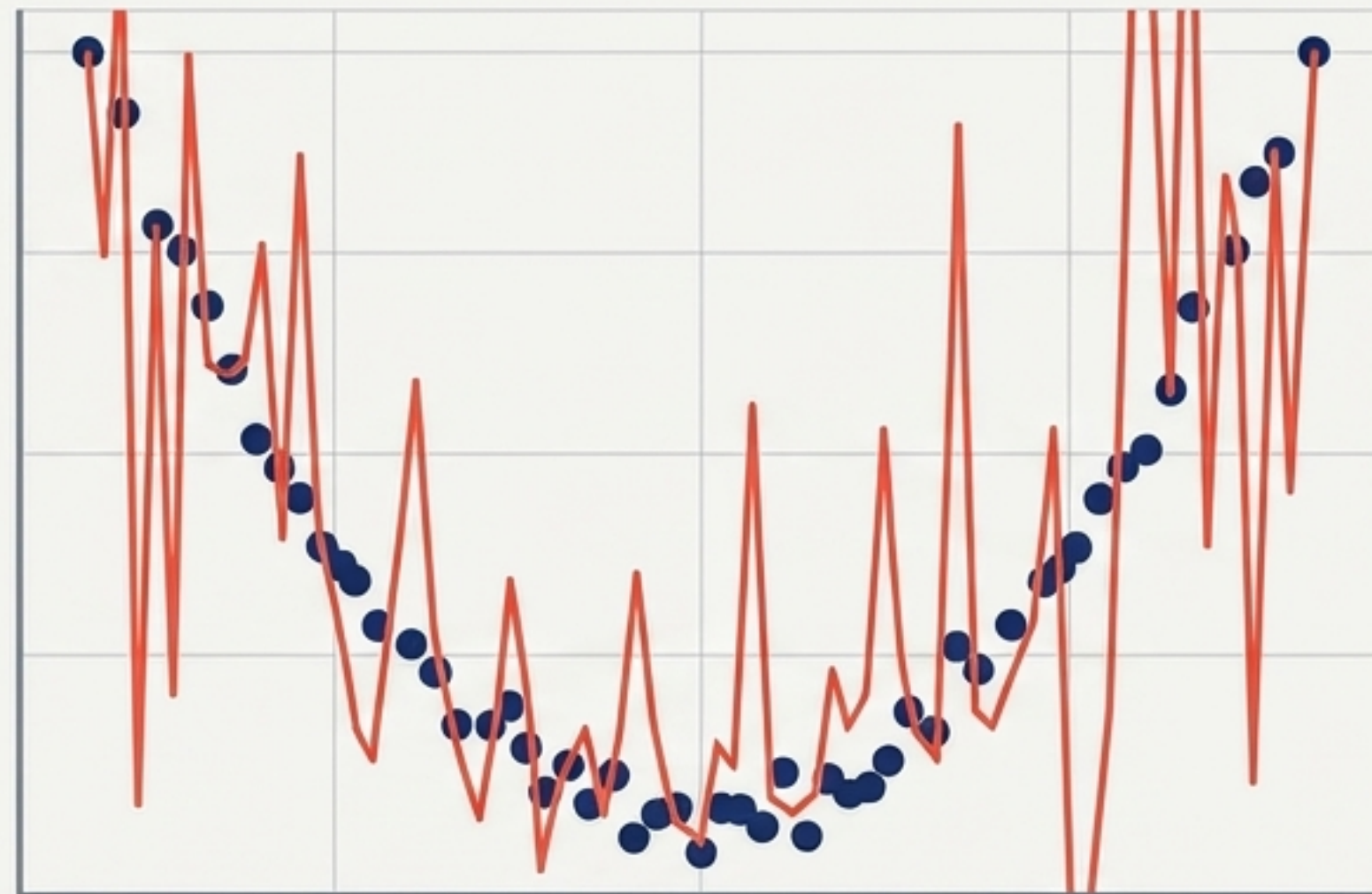
When the degree is too low, the model is too rigid. It completely misses the underlying mathematical patterns in both the training data and any future test data.

Overfitting: The Memorisation Trap

Degree 15



Degree 300



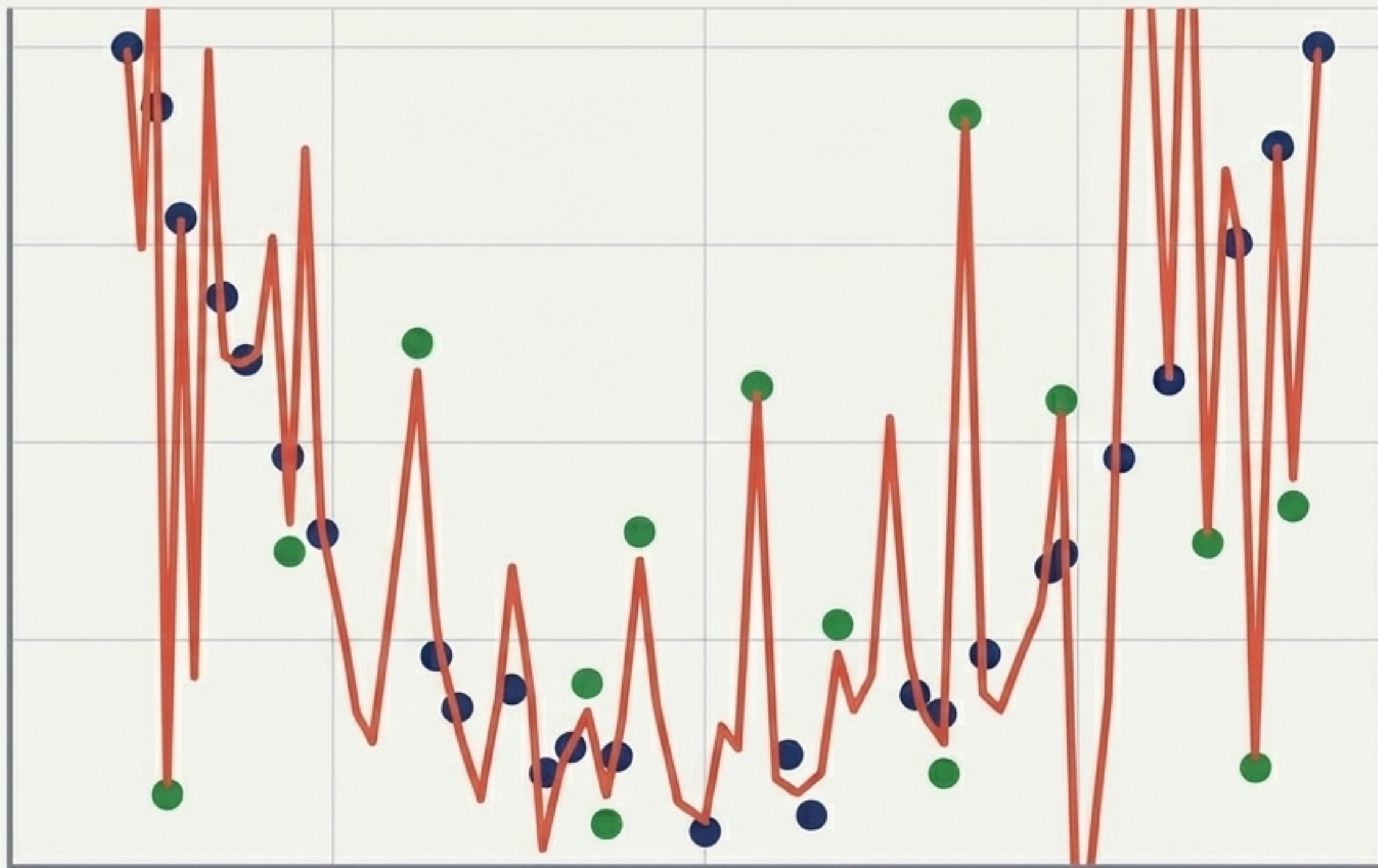
When the degree is pushed too high, the model becomes hypersensitive. Instead of learning the general trend, it memorises random noise.

The True Essence vs. The Noise

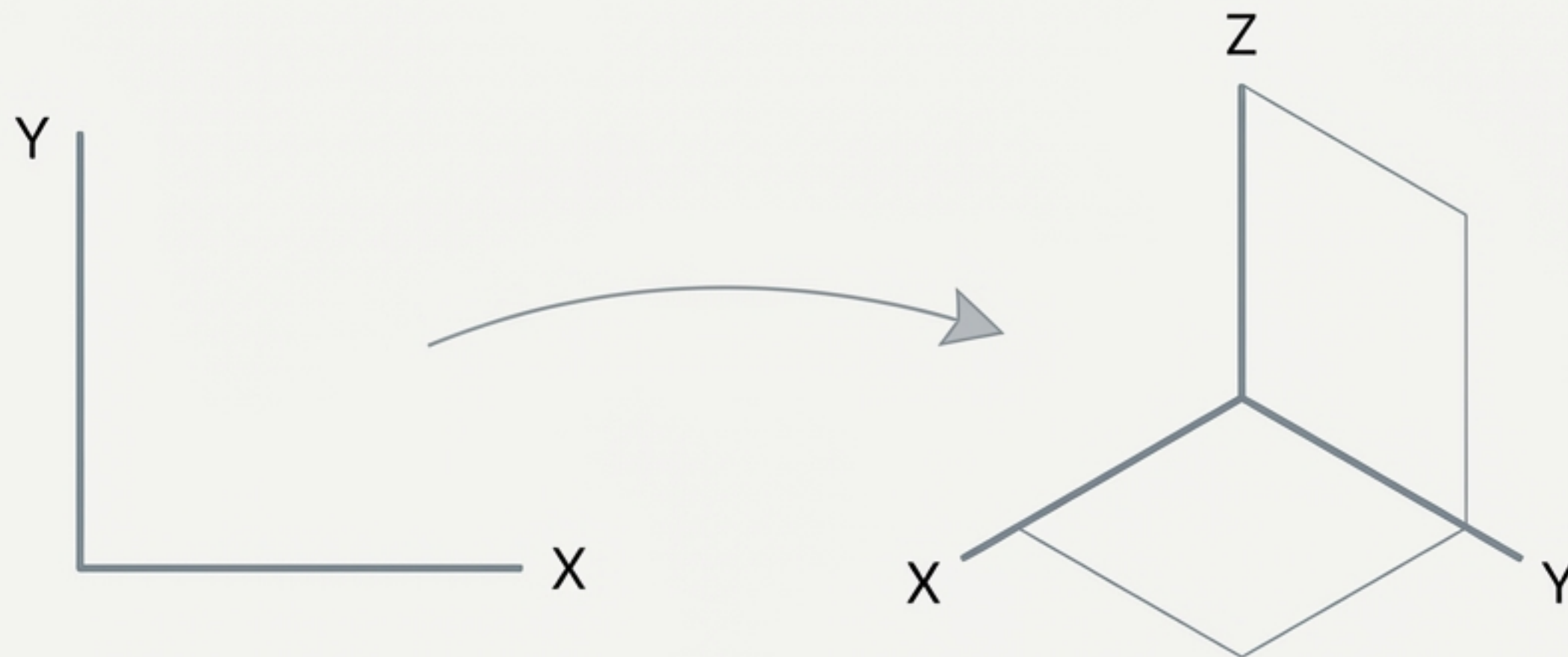
vs. The Noise

Blue = Training Data
Green = Test Data

A model that memorises noise fails entirely when exposed to new data. The goal is to capture the shape of the forest, not to map the exact location of every single leaf.

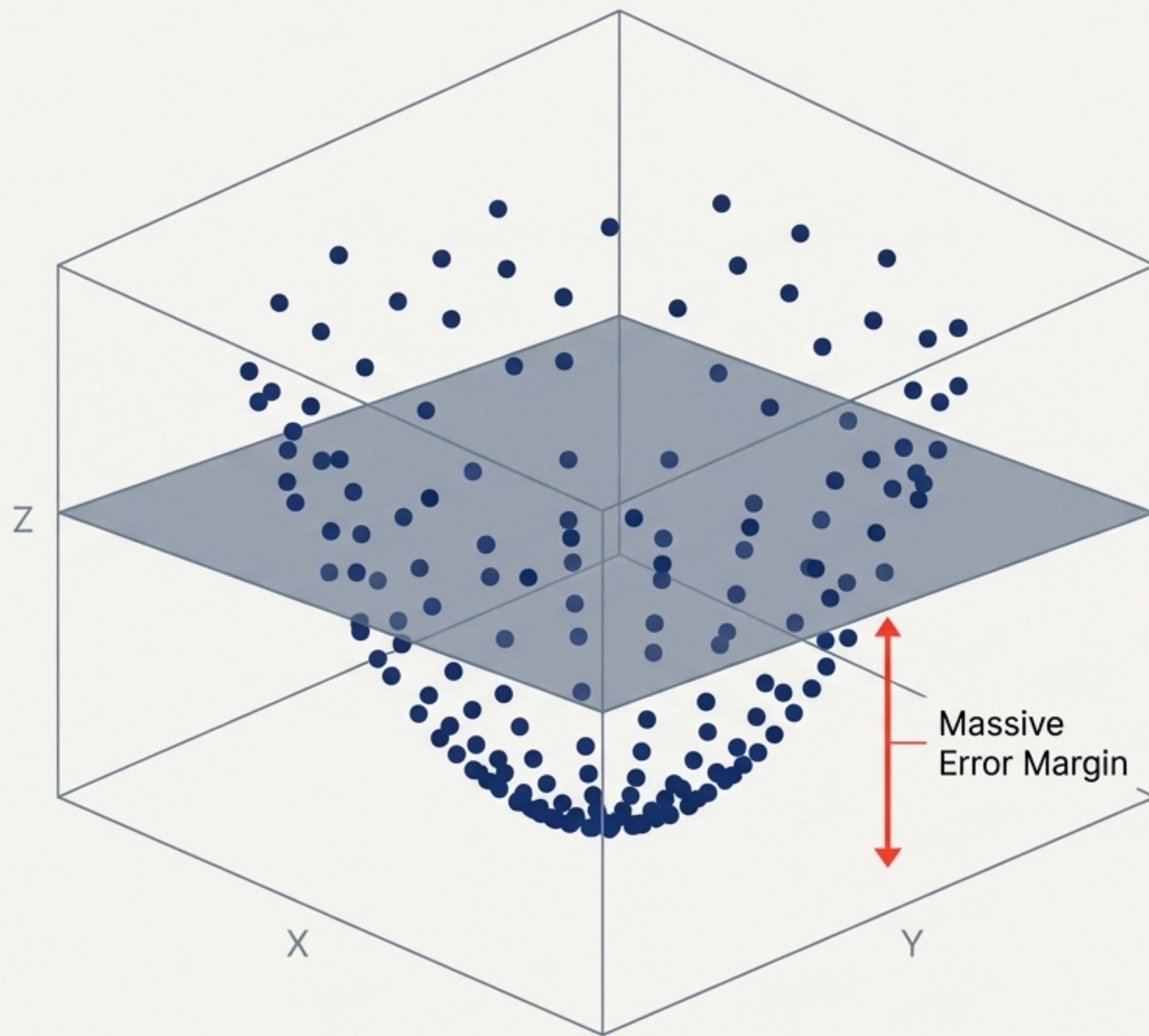


Scaling Up: Multiple Dimensions



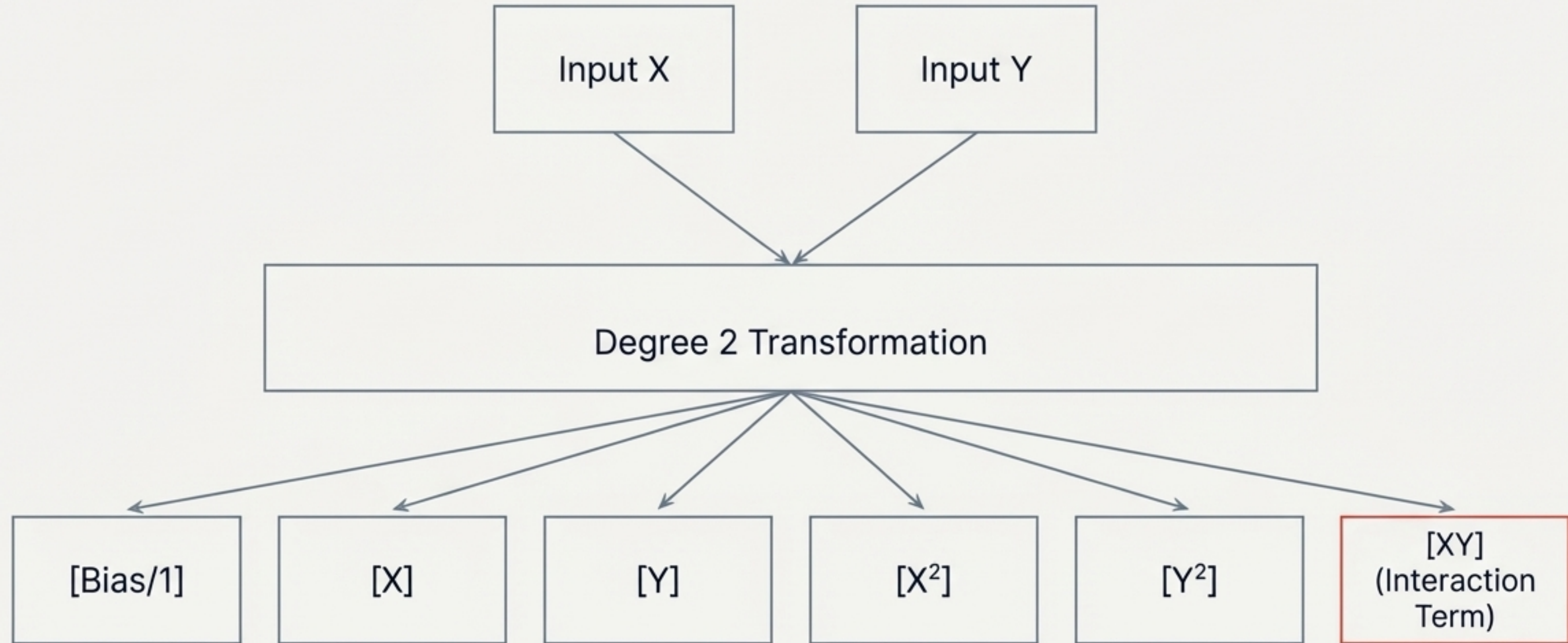
Real-world data rarely relies on a single input. What happens when an outcome (Z) relies on two separate, interacting inputs (X and Y)? We must move from simple lines to three-dimensional surfaces.

The 3D Problem



Standard Multiple Linear Regression attempts to fit a flat sheet of glass through a curved world. The errors are immense.

Feature Generation in Multi-Dimensions



The polynomial transformation generates isolated power terms for each variable, but crucially, it also multiplies them together to capture the interaction between different inputs.

The Math of Combinations

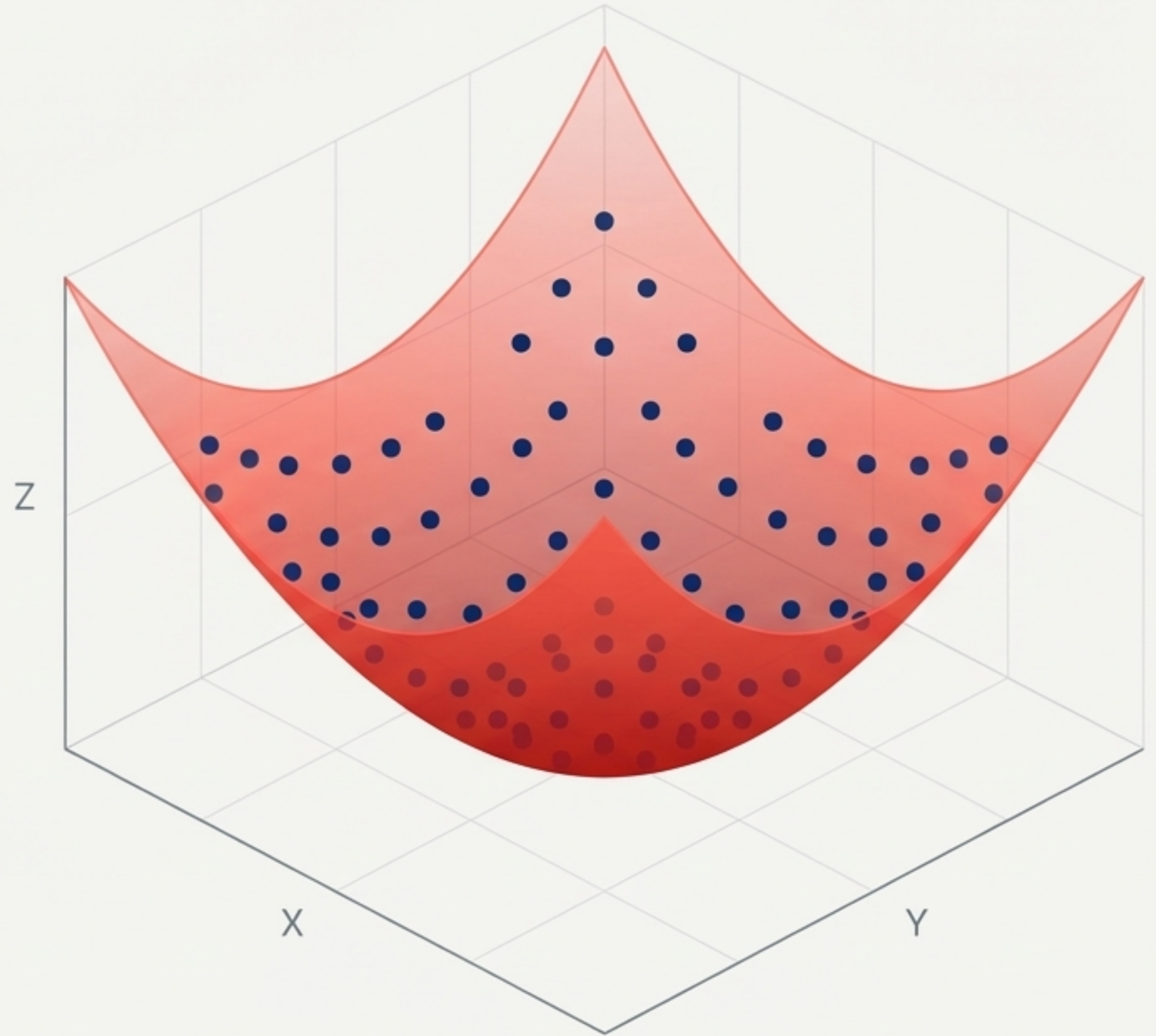
$$\del{X^2 \times Y^2 = \text{Power of 4}}$$

$$X^1 \times Y^1 = \text{Power of 2} \checkmark$$

Why does a Degree 2 model exclude terms like X^2Y^2 ? Because the sum of the combined powers (2+2) equals 4, which exceeds our specified degree limit. The maximum combined power of any interaction term can never exceed the overall model degree.

The 3D Solution

By applying Degree 2 **Polynomial Regression**, the model generates a curved surface that effortlessly maps the non-linear terrain.



(But beware: pushing the degree to 30 here will warp this elegant bowl into an unrecognisable, overfitting spike-trap.)